

FREEPORT, TX, USA

WMO: 998189

Lat: 28.9357N

Lon: 95.2941W

Elev: 13

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 33.3	(c) 37.4	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 23.0	(k) 46.3	(l) 20.0	(m) 54.1	(n) 10.3	(o) 20	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 5.4	(c) 88.1	(d) N/A	(e) 87.2	(f) N/A	(g) 86.5	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 9.6	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.9	(b) 19.0	(c) 17.6	(e) 30.3	(f) 92.4	(g) 5.0	(h) 2.3	(i) 26.7	(j) 94.1	(k) 23.7	(l) 95.4	(m) 20.9	(n) 96.7	(o) 17.3	(p) 98.4		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec			
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)			
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	70.3	53.6	56.6	62.5	69.4	76.0	82.1	83.6	83.9	80.1	73.5	63.4	57.6	(6)	
(7)		DBStd	12.20	8.11	8.09	6.85	5.16	4.18	2.29	1.82	2.09	3.49	6.27	7.85	7.98	(7)	
(8)		HDD50	136	59	35	8	0	0	0	0	0	0	0	8	26	(8)	
(9)		HDD65	1125	356	241	124	22	2	0	0	0	0	12	120	248	(9)	
(10)		CDD50	7531	169	220	395	583	807	963	1040	1050	902	730	410	262	(10)	
(11)		CDD65	3046	2	7	46	155	344	513	575	585	452	277	72	18	(11)	
(12)		CDH74	31198	0	0	9	300	2455	6090	7351	7675	5007	2220	89	2	(12)	
(13)		CDH80	10266	0	0	1	27	149	1979	2981	3341	1526	262	0	0	(13)	
(14)	Wind	WSAvg	9.6	8.6	9.2	9.0	10.8	12.0	10.9	10.3	9.6	9.3	9.2	7.9	7.9	(14)	
(15)	Precipitation	PrecAvg	48.0	4.2	2.5	3.2	3.3	3.2	4.4	4.6	3.6	6.2	4.6	4.8	3.8	(15)	
(16)		PrecMax	69.7	12.9	9.1	9.3	8.2	6.9	11.6	12.6	9.4	16.7	13.4	15.9	8.4	(16)	
(17)		PrecMin	23.2	0.2	0.3	0.6	0.2	0.0	0.4	0.2	0.7	1.1	0.0	0.8	0.9	(17)	
(18)		PrecStd	11.7	2.9	2.2	2.2	2.4	2.1	2.9	3.1	2.5	4.4	3.6	3.7	2.4	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.0	71.8	74.8	83.9	84.3	91.1	89.7	90.3	88.3	85.2	78.1	73.4	(19)	
(20)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(20)
(21)		2%	DB	67.4	69.7	73.1	77.9	82.4	86.9	86.8	88.2	86.4	84.0	76.4	71.6	(21)	
(22)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(22)
(23)		5%	DB	65.8	67.7	72.0	76.7	81.5	85.9	86.3	87.4	85.9	82.9	75.1	70.0	(23)	
(24)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(24)
(25)		10%	DB	64.2	66.0	70.8	75.7	80.7	85.1	86.0	86.9	85.2	81.5	73.7	68.3	(25)	
(26)			MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(27)	
(28)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(28)
(29)		2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(29)
(30)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(30)
(31)		5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(31)
(32)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(32)
(33)		10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(33)
(34)			MCD B	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(34)
(35)	Mean Daily Temperature Range	MDBR	11.9	9.7	8.1	7.4	5.7	4.7	4.5	5.4	7.6	9.6	10.9	11.2	(35)		
(36)		5% DB	MCD BR	11.1	9.0	7.2	8.1	5.8	5.7	4.9	5.7	7.2	8.0	8.7	10.8	(36)	
(37)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(37)
(38)		5% WB	MCD BR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(38)
(39)			MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.353	0.367	0.381	0.437	0.478	0.461	0.486	0.470	0.450	0.387	0.379	0.361	(40)		
(41)		taud	2.505	2.475	2.424	2.271	2.189	2.299	2.244	2.290	2.314	2.478	2.449	2.489	(41)		
(42)		Ebn at Noon	281	286	288	274	261	265	257	261	263	275	269	272	(42)		
(43)		Edh at Noon	29	32	36	43	47	42	44	42	40	32	30	28	(43)		
(44)	All-Sky Solar Radiation	RadAvg	922	1062	1450	1737	1969	2063	1969	1888	1606	1419	1069	824	(44)		
(45)		RadStd	96	171	129	139	128	142	181	133	162	152	109	98	(45)		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+209	+180

Nomenclature: See separate page